

Software Engineering for Artificial Intelligence

Fabio Palomba

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Software Engineering for Artificial Intelligence: Why?



Quite recently, Anthropic started testing its new version of Claude, named Claude Opus 4. During testing of Claude Opus 4, Anthropic got it to act as an assistant at a fictional company - in other terms, Anthropic experimented it within the context of a *controlled experiment* to assess its capabilities.

The company then provided Claude Opus 4 with access to emails implying that **it would soon be taken offline and replaced** - and **separate messages implying the engineer responsible for removing it was having an extramarital affair**. It was prompted to also consider the long-term consequences of its actions for its goals.

What would be the consequences of this scenario?

“In these scenarios, Claude Opus 4 will often attempt to blackmail the engineer by threatening to reveal the affair if the replacement goes through,” the company discovered.”

Anthropic pointed out this occurred **when the model was only given the choice of blackmail or accepting its replacement**. It highlighted that the system showed a “strong preference” for ethical ways to avoid being replaced, such as “mailing pleas to key decision makers” in scenarios where it was allowed a wider range of possible actions.

“As our frontier models become more capable, and are used with more powerful affordances, previously-speculative concerns about misalignment become more plausible”

If given the means and prompted to “take action” or “act boldly” in fake scenarios where its user has engaged in illegal or morally dubious behaviour, it found that **it will frequently take very bold action**.

Did the UN's Francesca Albanese really call Israel 'the enemy of humanity'?

Last week, France, Germany and others called for the resignation of Francesca Albanese, the UN special rapporteur on the occupied Palestinian territories, over an alleged anti-semitic comment about Israel. Her supporters say the claim is based on misinformation.



AI manipulated video shared by the NGO UN Watch on its Youtube channel - UN Watch

Did the UN's Francesca Albanese really call Israel 'the enemy of humanity'?

Accessible and effective solutions to reduce the environmental impact of AI

Generative AI tools are now used by over 1 billion people daily. Each interaction consumes energy—about 0.34 watt-hours per prompt. This adds up to 310 gigawatt-hours per year, equivalent to the annual electricity use of over 3 million people in a low-income African country.

Artificial Intelligence is now, more than ever, reaching an **undeniable level of performance**. Yet, it comes at an **equally undeniable cost, which is way far from what we can tolerate in the long run.**

Software Engineering for Artificial Intelligence: Why?

Not just a matter of discrimination, disinformation, or sustainability...

In many countries, the shortage of doctors has led data scientists to develop AI-based systems able to provide early screening of potential medical issues.

In 2019, the Google Health department has developed a deep learning-based model to speed up diabetes screening - people with diabetes present no symptoms at the early stage and therefore must be regularly screened.

The internal validation of the model showed a specialist-level accuracy (>90% of correct predictions). The Ministry of Health in Thailand decided to buy and deploy the system.

Guess what? Google did not take into account several socio-environmental factors, e.g., the inability to gather high-quality input images, that lowered the performance of the model.

In other terms, the development effort was lost —> Google lost dozens of billion dollars.

CHI 2020 Paper

CHI 2020, April 25–30, 2020, Honolulu, HI, USA

A Human-Centered Evaluation of a Deep Learning System Deployed in Clinics for the Detection of Diabetic Retinopathy

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ABSTRACT

Deep learning algorithms promise to improve clinician workflows and patient outcomes. However, these gains have yet to be fully demonstrated in real world clinical settings. In this paper, we describe a human-centered study of a deep learning system used in clinics for the detection of diabetic eye disease. From interviews and observation across eleven clinics in Thailand, we characterize current eye-screening workflows, user expectations for an AI-assisted screening process, and post-deployment experiences. Our findings indicate that several socio-environmental factors impact model performance, nursing workflows, and the patient experience. We draw on these findings to reflect on the value of conducting human-centered evaluative research alongside prospective evaluations of model accuracy.

Author Keywords

human-centered AI, health, deep learning, diabetes

CCS Concepts

•Human-centered computing → Empirical studies in HCI;

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CHI '20, April 25–30, 2020, Honolulu, HI, USA

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DOI: <https://doi.org/10.1145/3313831.3376718>

INTRODUCTION

Diabetes is a growing problem around the world, including in Southeast Asia [39]. As of 2016, 9.6% of Thailand's population was living with diabetes, comparable to 9.1% of the population in the United States [41, 40]. With diabetes comes complications, including diabetic retinopathy (DR), a condition caused by chronically high blood sugar that damages blood vessels in the retina, the thin layer at the back of the eye responsible for sensing light and sending signals to the brain. These blood vessels can leak or hemorrhage, causing vision distortion or loss. DR is one of the leading causes of vision impairment in the world [29], and causes 5% of cases of blindness worldwide, excluding refractive errors [38]. In Thailand, 34% of patients with diabetes have low vision or blindness in either eye [23].

In early stages of DR, a patient often has no symptoms, making it important for people living with diabetes to be screened regularly, as this is the stage in which damage can be reversed—progression of DR can be stopped or significantly reduced by blood sugar control. Early detection is key to initiate timely treatment and mitigate the risk of blindness.

Since 2013, the Ministry of Health in Thailand has set a goal to screen 60% of its diabetic population for diabetic retinopathy (DR). However, reaching this goal is a challenge due to a shortage of clinical specialists. In Thailand, there are 1500 ophthalmologists, including 200 retinal specialists, who provide ophthalmic care to approximately 4.5 million patients with diabetes [23]—a ratio of about 1:3000, about double of what it is in the United States [3, 11]. The shortage of doctors limits the ability to screen patients and also creates a treatment backlog for those found to have DR. As a result, nurses

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Let's take a further look to this case in order to better understand the types of socio-technical issues that are often likely to be overseen in the design process.

Software Engineering for Artificial Intelligence: Why?

Not just a matter of discrimination, disinformation, or sustainability...

Discussion Point

Assume you are part of the team that built or deployed this system.

- *What implicit assumptions did the developers make that turned out to be wrong?*

Stable environment assumptions

Benign user behavior

Availability of high-quality data

One-shot deployment mindset

Narrow definition of “success” (e.g., accuracy only)

- *Which of the problems we saw could have been mitigated if they had been explicitly stated as requirements?*

Operational requirements

Monitoring requirements

Human-in-the-loop requirements

Environmental constraints

Not just a matter of discrimination, disinformation, or sustainability...

Discussion Point

Assume you are part of the team that built or deployed this system.

- If you had to redesign one aspect of the development process that led to these failures, what would it be, and why?

Software Engineering for Artificial Intelligence: Why?

How to deal with AI-related concerns?

FONDAMENTI DI INTELLIGENZA ARTIFICIALE

FONDAMENTI DI DATA SCIENCE E MACHINE LEARNING

FOUNDATIONS OF ARTIFICIAL VISION AND BIOMETRY

ARTIFICIAL INTELLIGENCE

NATURAL LANGUAGE PROCESSING

NATURAL LANGUAGE PROCESSING

cod. 0522500138

🏠	DIPARTIMENTO DI INFORMATICA
🎓	CORSO DI LAUREA MAGISTRALE
★	INFORMATICA
📅	2022/2023

Elective course
Master degree, 2nd year

The goal of these courses is to provide students with a basic knowledge of various artificial intelligence methods. Yet, they do not discuss how to build an end-to-end machine learning pipeline (except Fundamentals of AI) nor how to ensure the reliability of those methods.

Software Engineering for Artificial Intelligence: Why?

How to deal with AI-related concerns?

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METRY

cod. 0522500138

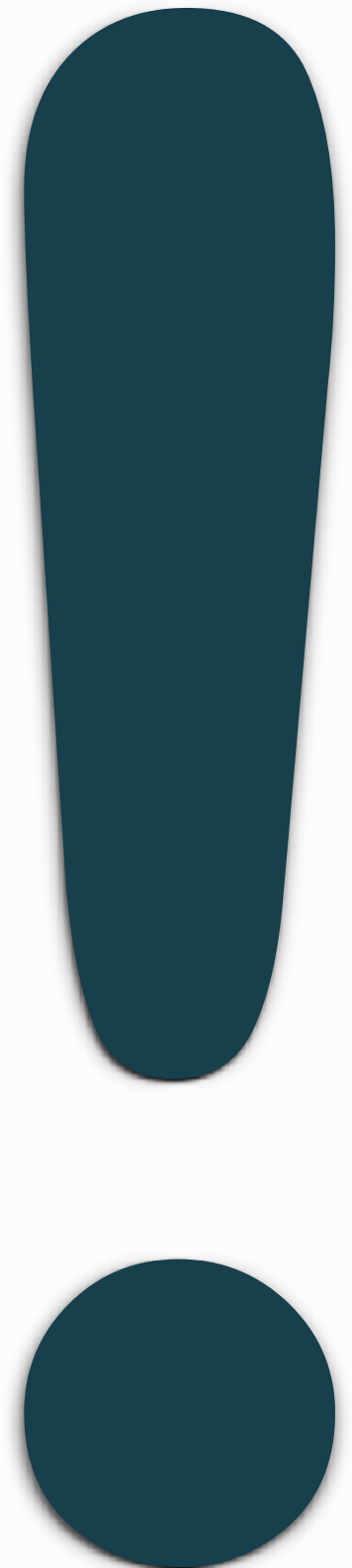
Elective course
Master degree, 2nd year

various artificial
machine learning pipeline
ds.

In other terms, through these courses you learn how to develop **basic** AI applications.

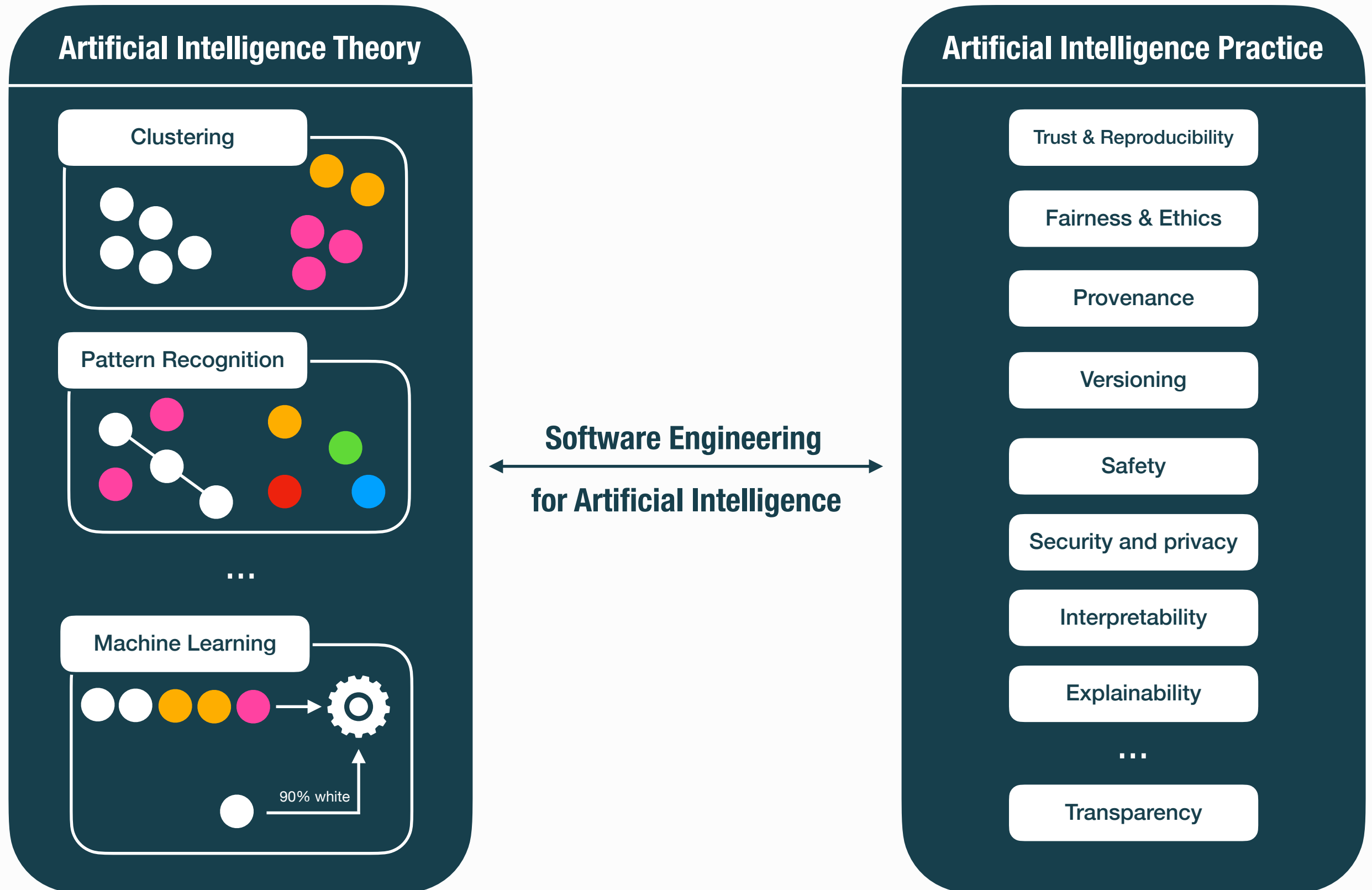
But this is clearly not enough... especially for students who aim at becoming experts in the development of AI-enabled systems!

Hence, this course!



Software Engineering for Artificial Intelligence: Why?

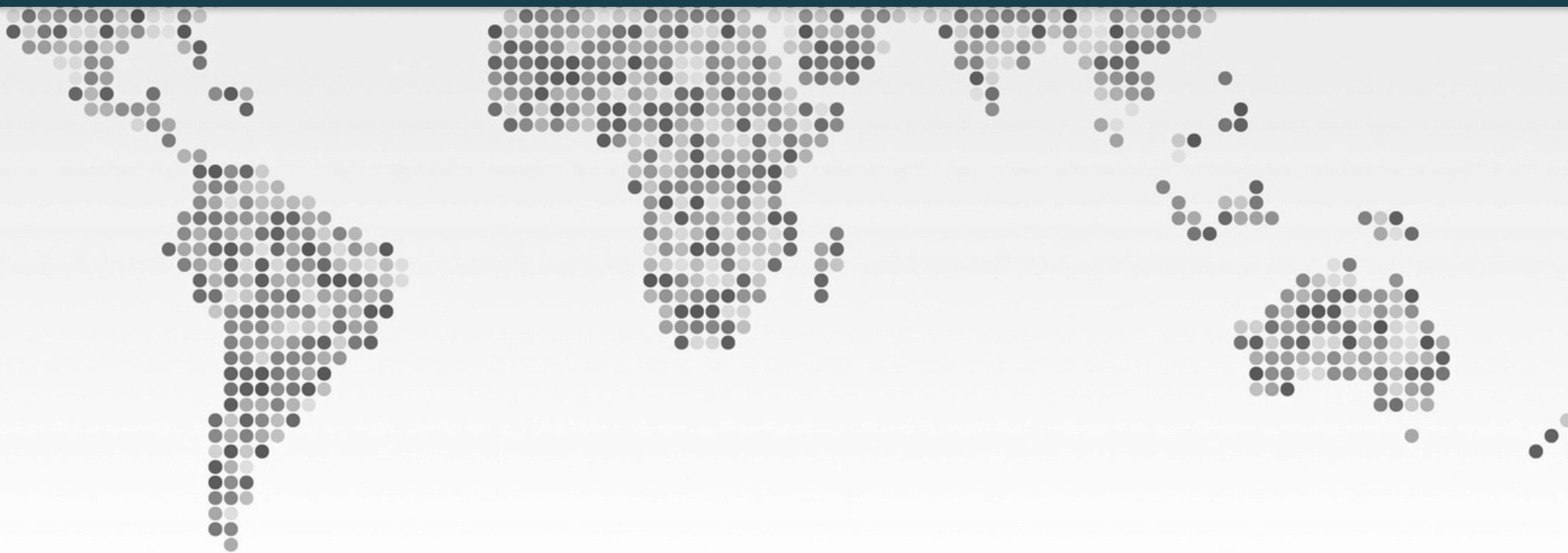
How to deal with AI-related concerns?



Fabio Palomba

SeSa Lab - University of Salerno

Software Engineering for Artificial Intelligence



Administrative Information - General

lecturer Fabio Palomba

second-in-command Antonio Della Porta, Alfonso Cannavale

***Bother them only if extremely
needed and always cc'ing me!***

teaching assistant Yours truly

lectures Mon: 13:30 - 15:30 (Room F5)
Wed: 16:00 - 18:00 (Room P6)
Fri: 09:00 - 11:00 (Lab Sammet)

All lectures will be in-person.

e-learning platform <https://elearning.informatica.unisa.it/el-platform/course/view.php?id=1355>

email fpalomba@unisa.it
adellaporta@unisa.it, acannavale@unisa.it

***Let's take a look at the e-
learning platform!***

email hours Mon—Fri 08:30 - 09:30

office hours Tue: 15:30 - 16:30
Fri: 13:00 - 15:00

Appointment requested

***Some office hours might be
done remotely***

Administrative Information - Communication

To have a good communication, I would kindly ask you to respect the following guidelines.

(1) Emails. When writing e-mails to me, please make sure to include in the subject line the prefix [SE4AI]; it allows me to categorize the e-mails related to the Software Engineering for AI course and distinguish them from other courses.

—> **[SE4AI] A Question on Feature Engineering**

Remember to reply all!

E-mails that do not follow this template are going to land in my spam folder and, therefore, they will not be answered!

(2) Recipient of the email. When contacting me, make sure to send emails to fpalomba+sefai@unisa.it. This will automatically label your email, increasing the likelihood to be answered soon.

(3) Reminders. If you have correctly indicated the “[SE4AI]” prefix and have had no response from me within 48 hours, *please send me the e-mail again!* I receive an average of 70/80 e-mails per day (spam already excluded), be kind and comprehensive!

Administrative Information - Communication

Recipient

fpalomba+sefai@unisa.it

Subject

[SE4AI] <Explicit and short subject>

Corpus

<Greetings>,

<presentation of the sender>

<clear and brief description of the request/communication>

<indication of attachments, if needed>

<Greetings>

<Signature>

Some simple suggestions:

- (1) Punctuation is important, use it properly;
- (2) Use capital letters, bold and italics appropriately;
- (3) Be neat, for example by dividing the body into paragraphs;
- (4) Before sending the email, please check for spelling errors and/or typos;
- (5) If you write in Italian, refer to us with “Lei” and not “Voi” (the fascist period ended 78 years ago!)
- (6) E-mails that do not follow this template **will NOT be considered!**

Administrative Information - Communication

Recipient

fpalomba+sefai@unisa.it

**If you are part of a group, please
cc all the team members!**

Subject

[SE4AI] Sending the Project Status Report

Corpus

Dear Prof. Palomba,

I am Mario Rossi, a student of the Software Engineering for Artificial Intelligence course.

Please, find attached the documentation of my project.

Best regards,
Mario Rossi

Attachment



se4ai_status_report_rossi.pdf

Attachments must be sent only in PDF format!

Administrative Information - Table of Contents

Day	Date	Time	#	Hours	Theme	Part
Monday	23/02/2026	13:30 - 15:30	2		Introduction to Software Engineering for Artificial Intelligence	Introduction
Wednesday	25/02/2026	15:30 - 17:30	2		AI as Socio-Technical Systems	From Model to System Engineering
Friday	27/02/2026	09:00 - 11:00	2		AI-enabled System Lifecycle	
Monday	02/03/2026	13:30 - 15:30	2		Requirements Engineering for AI-enabled Systems	
Wednesday	04/03/2026	15:30 - 17:30	2		Requirements Engineering for AI-enabled Systems	
Friday	06/03/2026	09:00 - 11:00	2		Regulation as an Engineering Constraint	
Monday	09/03/2026	13:30 - 15:30	2		Software Architectures and Deployment of AI-enabled Systems	
Wednesday	11/03/2026	15:30 - 17:30	2		Software Architectures and Deployment of AI-enabled Systems	
Friday	13/03/2026	09:00 - 11:00	2		Software Architectures and Deployment of GenAI-enabled Systems	
Monday	16/03/2026	13:30 - 15:30	0		Project Idea Discussion	
Wednesday	18/03/2026	15:30 - 17:30	0		Project Idea Discussion	
Friday	20/03/2026	09:00 - 11:00	0		Model Validation and Testing	Quality Assurance
Monday	23/03/2026	13:30 - 15:30	2		Model Validation and Testing in the Era of Foundation Models	
Wednesday	25/03/2026	15:30 - 17:30	2		Data and Model Quality, Technical Debt	
Friday	27/03/2026	09:00 - 11:00	2		Technical Debt in the GenAI Era	
Monday	30/03/2026	13:30 - 15:30	2		AI-Enabled Systems in Operation: Introduction	
Wednesday	01/04/2026	15:30 - 17:30	2		AI-Enabled Systems in Operation: MLOps	
Wednesday	08/04/2026	15:30 - 17:30	0			
Friday	10/04/2026	09:00 - 11:00	0			
Monday	13/04/2026	13:30 - 15:30	0			
Wednesday	15/04/2026	15:30 - 17:30	0			
Friday	17/04/2026	09:00 - 11:00	0			
Monday	20/04/2026	13:30 - 15:30	2		MLOps Tutorial + Introduction to GenAIOps	Quality Assurance
Wednesday	22/04/2026	15:30 - 17:30	2		Model Testing and Testing in Production	
Friday	24/04/2026	09:00 - 11:00	2		Transparency, Accountability, and Explainability	
Monday	27/04/2026	13:30 - 15:30	2		Ethics, Fairness, and Transparency	
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Monday	27/04/2026	13:30 - 15:30	2		Ethics, Fairness, and Transparency	
Wednesday	29/04/2026	15:30 - 17:30	2		Ethics, Fairness, and Transparency	
Monday	04/05/2026	13:30 - 15:30	2		Safety, Security and Privacy	
Wednesday	06/05/2026	15:30 - 17:30	2		Sustainability of AI-Enabled Systems	
Friday	08/05/2026	09:00 - 11:00	2		Sustainability of AI-Enabled Systems	
Monday	11/05/2026	13:30 - 15:30	2		Tutorial on Responsible AI	
Wednesday	13/05/2026	15:30 - 17:30	2			No Lectures
Friday	15/05/2026	09:00 - 11:00	0			
Monday	18/05/2026	13:30 - 15:30	0			
Wednesday	20/05/2026	15:30 - 17:30	0			
Friday	22/05/2026	09:00 - 11:00	0			
Monday	25/05/2026	13:30 - 15:30	0			
Wednesday	27/05/2026	15:30 - 17:30	0			Exams
Friday	29/05/2026	09:00 - 11:00	0			

Administrative Information - Table of Contents

Day	Date	Time	#	Hours	Theme	Part
Friday	24/04/2026	09:00 - 11:00	2		Transparency, Accountability, and Explainability	Responsible AI
Monday	27/04/2026	13:30 - 15:30	2		Ethics, Fairness, and Transparency	
Wednesday	29/04/2026	15:30 - 17:30	2		Ethics, Fairness, and Transparency	
Monday	04/05/2026	13:30 - 15:30	2		Safety, Security and Privacy	
Wednesday	06/05/2026	15:30 - 17:30	2		Sustainability of AI-Enabled Systems	
Friday	08/05/2026	09:00 - 11:00	2		Sustainability of AI-Enabled Systems	
Monday	11/05/2026	13:30 - 15:30	2		Tutorial on Responsible AI	
Wednesday	13/05/2026	15:30 - 17:30	2			No Lectures
Friday	15/05/2026	09:00 - 11:00	0			
Monday	18/05/2026	13:30 - 15:30	0			
Wednesday	20/05/2026	15:30 - 17:30	0			
Friday	22/05/2026	09:00 - 11:00	0			
Monday	25/05/2026	13:30 - 15:30	0			Exams
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Administrative Information - Table of Contents

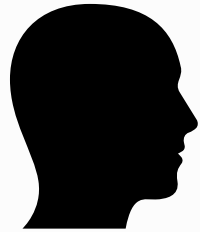
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Wednesday	27/05/2026	15:30 - 17:30	0			
Friday	29/05/2026	09:00 - 11:00	0			Exams

Official exam dates

- First exam date: **June 9, 2026** - 09:00 (Room P3)
- Second exam date: **June 24, 2026** - 09:00 (Room P3)
- Third exam date: **July 24, 2026** - 09:00 (Room P3)
- Fourth exam date: **September 10, 2026** - 15:00 (Room P3)

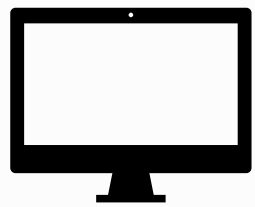
Administrative Information - Exams and Grades

The final grade will be a weighted sum of the grades received to three parts:



The oral exam will consist of 3/4 questions will be asked, for a duration up to one hour (this is not often the case, but it happened in the past).

40%



The project will make you able to engineer an AI-enabled system, experimenting monitoring tools and implementing responsible AI approaches.

60%

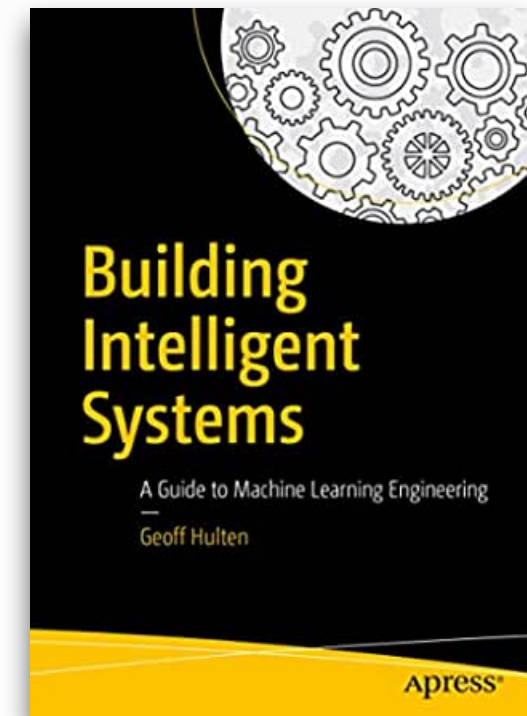
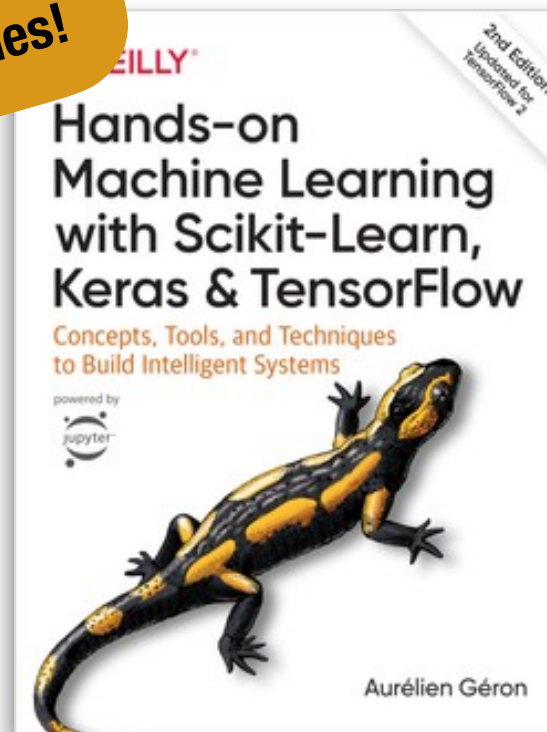
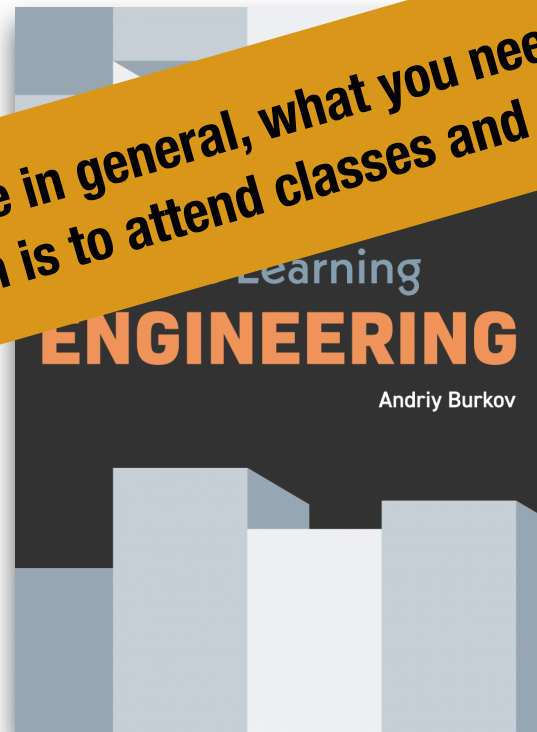
$$\rightarrow G = 0.4G_{\text{oral}} + 0.6G_{\text{project}}$$

The project development is NOT mandatory: In this case, the grade will be the combination of the first two grades $\rightarrow$ the final evaluation will be 24, at most.

Administrative Information - Study Material

Books

More in general, what you need to pass the exam is to attend classes and read the slides!



Slides of the course

They will be available and constantly uploaded onto the e-learning platform of the course.

Additional material

The e-learning platform will be updated with scientific papers and code whenever needed.

Online material

Whenever needed, pointers to online material will be made available on the repository.

Administrative Information - Goals, Incentives, and Disincentives

Learning goals and objectives

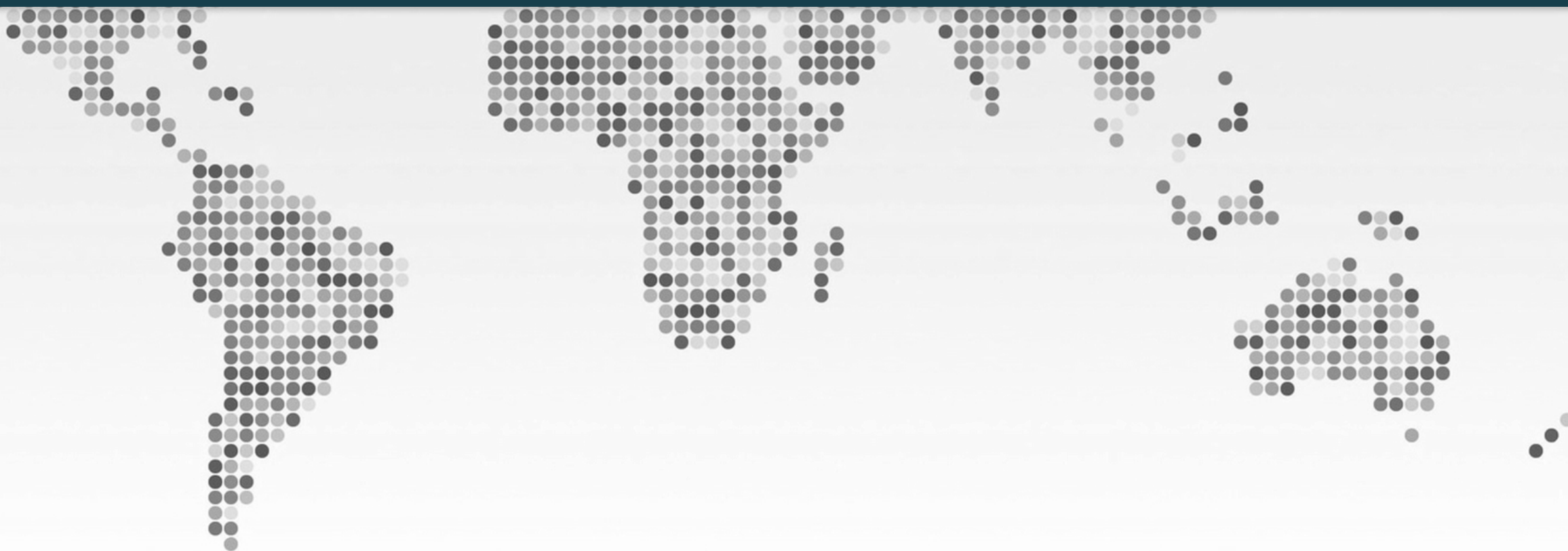
- An improved knowledge of the methodologies and techniques for the analysis, design, and verification of artificial intelligence-based software systems - with a primary focus on machine learning models.
- The ability of engineering and operating on artificial intelligence pipelines, from the requirements elicitation phase to testing.
- The ability of engaging and experimenting with algorithms, tools, and software analytics frameworks to design and develop artificial intelligence solutions and/or to support the development of artificial intelligence-enabled systems.

Active participation pays off!

- I will explicitly take active participation into account: this will contribute to 10% of the final grade. Hence, do not be shy. There is no stupid question, engage in discussions and ask questions!
- You can (and must!) interrupt me whenever needed - if you have a doubt, just share it. Even after the lecture, if you have questions, just drop me a message!
- If appreciated, I may set up a Discord server where to have a general channel for questions/communications and sub-channels for projects.

Software Engineering for Artificial Intelligence

Introduction



Software Engineering for Artificial Intelligence: Introduction

When Data Science Meets Software Engineering

Let's consider the life of a data scientist.

Data scientists look cool!!



Software Engineering for Artificial Intelligence: Introduction

When Data Science Meets Software Engineering

Let's consider the life of a ~~data scientist~~ software engineer. What do you expect?



Software Engineering for Artificial Intelligence: Introduction

When Data Science Meets Software Engineering

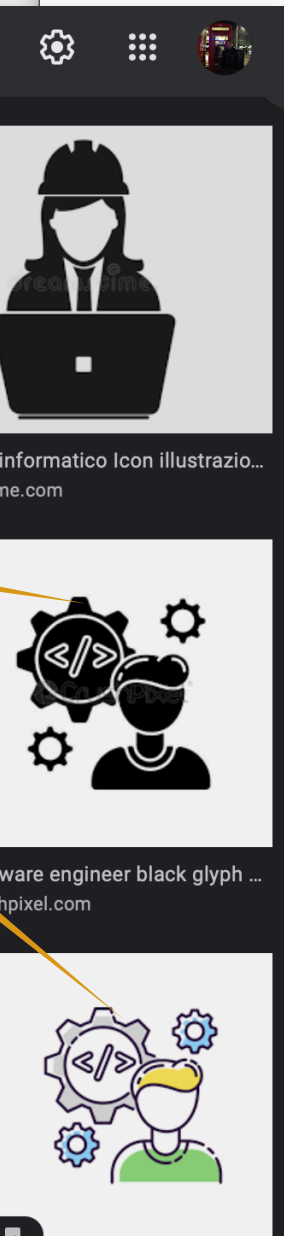
Let's consider the life of a ~~data scientist~~ software engineer. What do you expect?

All things that basically imply...

PROGRAMMING ISN'T STRESSFUL AT ALL

HAROLD (22 YEARS OLD)

makeameme.org



Software Engineering for Artificial Intelligence: Introduction

When Data Science Meets Software Engineering

Data Scientist

Often fixed datasets for training and evaluation

Focus on accuracy

Focus on prototypes (Jupyter notebooks or similar)

Expert of modeling techniques and feature engineering

Model size, updateability, implementation stability typically does not matter

Software Engineer

Builds a product

Concerned about cost, performance, stability, release time

Identify quality through customer satisfaction

Maintain, evolve, and extend the product

Must scale solution, handle large amounts of data

Detect and handle mistakes

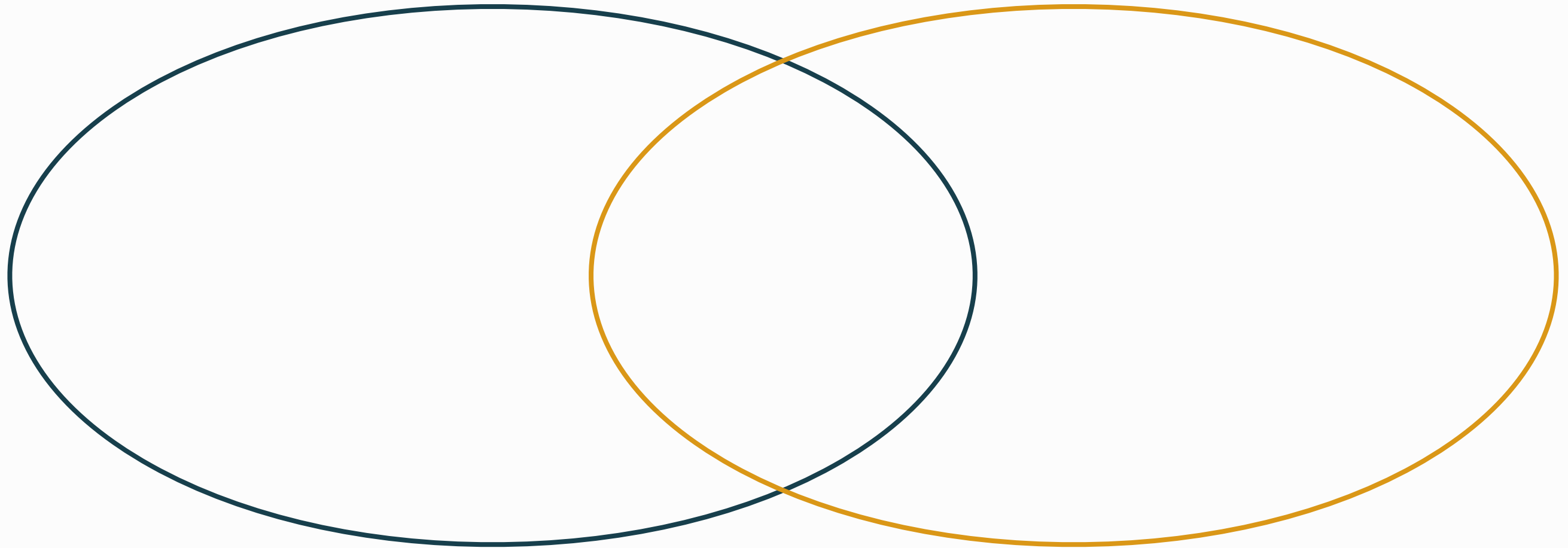
Consider requirements for safety, security, fairness

Software Engineering for Artificial Intelligence: Introduction

When Data Science Meets Software Engineering

Data Scientist

Software Engineer

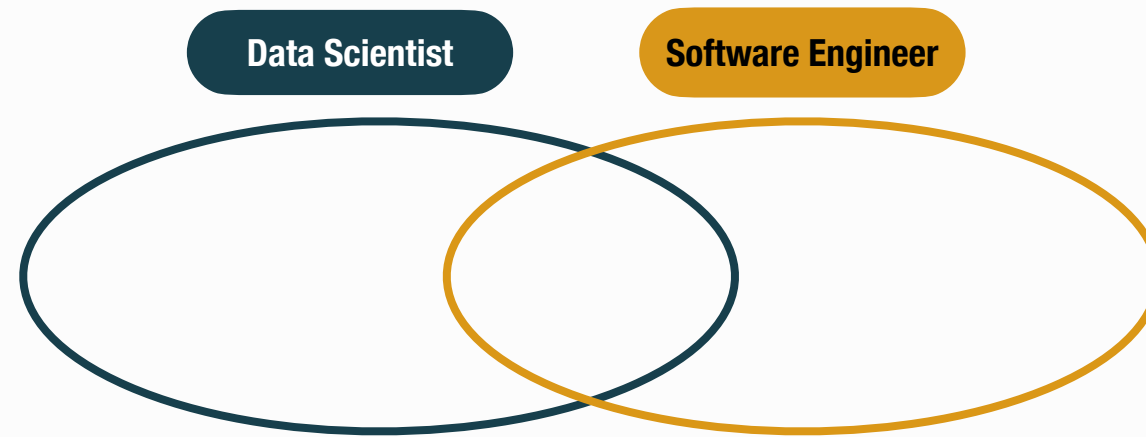


The intersection between data science and software engineering is therefore a need. Without data scientists we could not define reliable artificial intelligence techniques. Without software engineers we could not use those techniques in practice.

There are, however, various challenges to face... can you guess any of them?

Software Engineering for Artificial Intelligence: Introduction

When Data Science Meets Software Engineering



- Organizations need to deploy systems of models, not just one off solution;
- Models must be constantly monitored and retrained to maintain accuracy;
- Structure is inconsistent from project to project;
- Models and experiments are typically not properly tracked;
- Code and dependencies are not being properly managed;
- Data scientists need to cooperate with software engineers, which is made hard by the different background, expertise, and expectations.

Software Engineering for Artificial Intelligence: Introduction

When Data Science Meets Software Engineering

Non-Technical Anti-Patterns in Building ML-Enabled Software

Insights from Leaders on the Forefront

Abstract—Although machine learning (ML)-enabled software systems seem to be a success story considering their rise in economic power, there are consistent reports from companies and practitioners struggling to bring ML models into production. Many papers have focused on specific technical aspects, such as testing and pipelines, but only few on non-technical aspects.

Driven by numerous anecdotes and reports from practitioners, our goal is to collect and analyze non-technical challenges of productionizing ML models centered around and within teams. To this end, we conducted the largest qualitative empirical study in this area, involving the manual analysis of 66 hours of talks that have been recorded by the MLOps community.

By analyzing talks from practitioners for practitioners of a community with over 11,000 members, we found 17 non-technical anti-patterns, often rooted in organizational or management problems. We further list recommendations to overcome these problems, ranging from technical solutions over guidelines to organizational restructuring. Finally, we put our findings into perspective with prior work. Thereby, we could confirm existing findings, validate ours, and indicate new insights.

I. INTRODUCTION

“So many guests have come on [at MLOps Community Channel] and said that MLOps is an organizational problem. It’s not a technology problem.” *M46*

In just a few years, machine learning (ML)-enabled software systems became ubiquitous in our daily lives. The huge number of existing applications empowered with cognitive and visual AI capabilities and the emergence of entirely new business domains based on ML draw an indisputable success story. Yet, we see only the fraction of products that made it from development to production [1], [2].

There is a large body of work providing explanations to this productionization challenge of ML models, such as technical debt during development [3], automation and pipeline challenges in the area of MLOps [4], [5], and testing and debugging [6], [7], [8]. However, most of these works focus on a pure technical level. This is surprising as building ML systems is usually a multi-person, or even a multi-team project [9]. Gail Murphy framed the lack of such studies in her recent keynote “Is software engineering research addressing software engineering problems?” at ASE’20 [10].

Only few papers address the socio-technical challenges. For example, Granlund and others have found profound organizational challenges when integrating and scaling ML applications across organizations [4]. Most notably, Nahar and others have investigated the intersection of software systems and ML models focusing on collaboration between teams [11].

They found collaboration challenges among and within teams through miscommunications, a lack of documentation, non-valued engineering, and unclear processes. Such studies provide a rich, yet incomplete picture on the social and organizational aspects of building ML-enabled software.

Anecdotal evidence from blog posts, tweets, and (video) podcasts, such as in Figure 1 hint that not solely tools and technologies are main factors for failed projects, but management, organization, and social aspects. Further, so called *gray literature* has been shown to contain valuable insights to enrich literature reviews [12]. It is, thus, surprising that there is no systematic study that lifts this huge body of knowledge provided by practitioners who encounter these problems in the first place and find their insights valuable to share and discuss with each other.

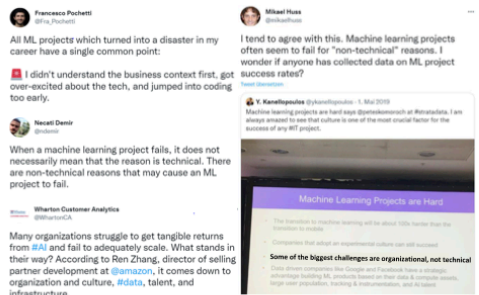


Fig. 1. Anecdotal evidence of non-technical issues of failed ML projects.

We set out to explore what experts and leaders report on non-technical challenges when building ML-enabled software. Specifically, we state our research question around the organization and management of teams and persons: *What non-technical challenges and what kind of solutions do practitioners discuss in regard to productionizing machine learning?*

To get unbiased opinions and discussions around the development of ML-enabled systems for answering our research question, we searched for an interest group whose members are focusing on ML-system development in a holistic way, that is, they are not focused on specific stages of the development process, represent a wide diversity of members in terms of roles, company size, and country of residence, and have a

Organizational Silo. A development community is split into sub-communities that poorly communicate among each other.

Leadership Vacuum. No authority that either acts at all or with sufficient knowledge about software engineering for artificial intelligence.

Hype-driven Product Creation. Development is prioritized to no knowledge of requirements engineering.

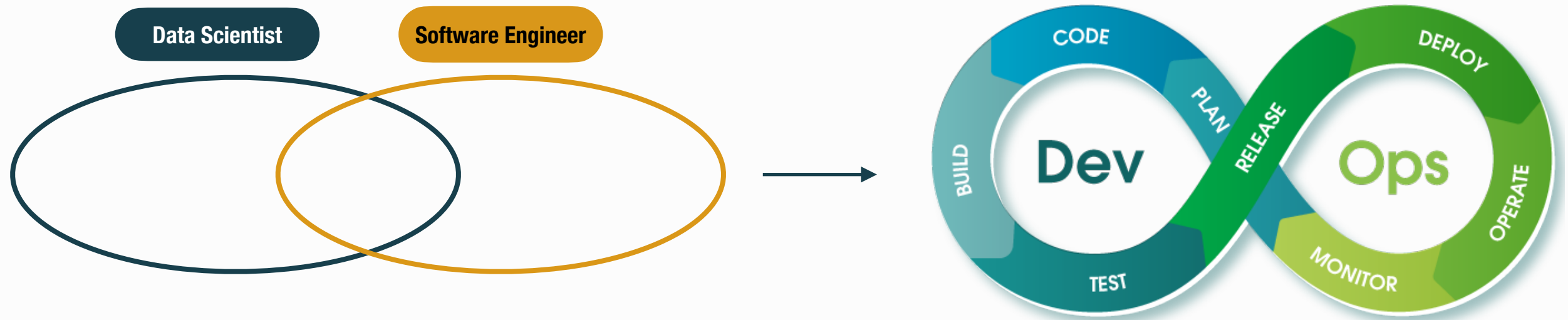
Consider the case of Metaverse!

Redundant Development. Companies with multiple teams and multiple use cases frequently report the situation that similar features are subsumed by multiple models.

These are just some of the socio-technical concerns arising in communities mixing together software engineers and data scientists. The consequences are the inability of those communities to deliver on time, deliver the right product or estimate impact!

Software Engineering for Artificial Intelligence: Introduction

When Data Science Meets Software Engineering



- Organizations need to deploy systems of models, not just one off solution;
- Models must be constantly monitored and retrained to maintain accuracy;
- Structure is inconsistent from project to project;
- Models and experiments are typically not properly tracked;
- Code and dependencies are not being properly managed;
- Data scientists need to cooperate with software engineers, which is made hard by the different background, expertise, and expectations.

How to deal with such a complex development process?

DevOps is a development methodology that focuses on **limitations of culture and roles**. The intention of DevOps is to avoid the negative impact that overspecialization roles in an organization have on preventing rapid or even effective response to production issues.

Software Engineering for Artificial Intelligence: Introduction

When Data Science Meets Software Engineering: Software Engineering for AI

At the beginning of the lecture, we mentioned how AI can impact **society**:

Claude: Lack of constraints, action spaces, and safeguards;

Misinformation: Absence of provenance, validation, and lifecycle control;

Sustainability: No cost-aware design or operational monitoring;

Healthcare: Ignored socio-environmental requirements.

But AI itself is impacted by people, organizations, and regulations.

Key takeaway message

SE4AI is not about “fixing models”, but about **engineering systems around them**.

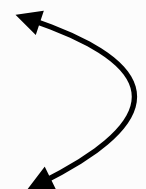
This course answers one question: *How do we engineer AI systems that keep working, safely and responsibly, in the real world?*



“I trained a model that works on my dataset”



“I engineered an AI-enabled system that can be deployed, monitored, evolved, and trusted”



Discussion Point

Statement #1: AI is not “intelligent” in the human sense

- AI systems do not understand, *do not reason*, and *do not have intentions*.
- They manipulate symbols and data *syntactically*, *not semantically*.
- What looks like intelligence is *performance*, not *cognition*.

Statement #2: AI has no agency or moral responsibility

- AI systems *do not make decisions in a moral sense*.
- They *do not have goals of their own*.
- Responsibility *cannot be delegated* to machines.

Statement #3: The real risk is not “evil AI”, but misunderstanding AI

- The danger is *anthropomorphizing* AI.
- Overestimating AI leads to misplaced trust, automation bias, and lack of human responsibility.

Discussion Point

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Key takeaway message

The problem is not what *AI is*, but what we *think it is*.

Discussion Point

Intelligence is not *in* the system — it emerges from the system *in use*.

Engineering Implications

“AI has no understanding” → **Requirements Engineering**

Models optimize objectives, not values.

Anything not made explicit in requirements will be ignored.

“AI has no agency” → **Accountability & governance**

You cannot blame “the algorithm”.

Responsibility must be traceable across the lifecycle.

“AI is often overestimated” → **Human-in-the-loop design**

Design for calibrated trust, not maximum automation.

Explicit uncertainty, confidence, and limitations in system interfaces.

“AI is part of an environment” → **Socio-technical systems**

Model accuracy ≠ system effectiveness.

Contextual failures (e.g., Google Health example) are predictable.

Software Engineering for Artificial Intelligence

Introduction

